STOCK (INWARD & OUTWARD) REGISTER

Maximum _____ ARTICLE _____ Rate _____ 67

Minimum _____

Date	PARTICULARS	Folio	QUANTITY Receipts	Issued	Balance	Remarks

Cs 401 Final term

Firmware ⇒ software to control hardware in ROM

BIOS ⇒ Basic input out services.

BIOS tests hardware when computer is switched on.

Bootstrap ⇒ loading of OS from hard disk

BIOS ⇒ low level

OS ⇒ high level build on BIOS.

BIOS layer ⇒ Display char, clear screen etc

Service no. is given in AH register. while sub service no is in AL or BL.

In interrupts are arguments are given through registers

Ralph brown list ⇒ Comprehensive reference of interrupts

B8000 ⇒ very old vedio card.

PIC ⇒ programmable interrupts controller.
It has 8 input single
and 1 output signals.
(0–7)

IBM XT ⇒ 1 PIC (8 possible interrupts)

IBM AT ⇒ 2 PIC (16 .. ..)

IRQ ⇒ Interrupts requests ⇒ 8 input signals to PIC.
IRQ 0 to IRQ 7.

8259 ⇒ Technical no. of PIC

IRQ 0 ⇒ derived by a timer device.
(higest perority)

IRQ 1 ⇒ derived by keyboard by pressing keys.

IRQ 2 ⇒ Cascading Interrupt

IRQ 3 ⇒ Connected to serial port COM2

IRQ 4 ⇒ Connected to serial port COM1.

IRQ 5 ⇒ Used by sound/networkcard/modem

IRQ 6 ⇒ Used by floppy disk

IRQ 7 ⇒ Used by parallel port

Maximum
Minimum

Date	PARTICULARS	Folio	QUANTITY Receipts	Issued	Balance	Remarks

IRQ conflict ⇒ two devices request for same IRQ line.

IRQ to INT mapping (mapping of IRQ to specific interrupts)

IRQ0 to IRQ7 mapped to Int 8 to F

EOI ⇒ End of interrupt, sent at end of process

* for communication with peripheral devices processor uses I/O ports.
* total 65536 possible I/O ports.
* 16 lower bits of address bus are used for I/O ports.

In MOV ⇒ processor selects memory address space

In INt ⇒ " " peripheral address space.

IN and Out ⇒ may have byte or word.
* but byte form is mostly used.

Date PARTICULARS Folio Receipts

AL or AX => source reg in Out and
desti reg in IN

Port no.

20 and 21 => PIC (20 for control port
and 21 is interrupt mask register
60 to 64 => key board

378 => parallel port / data port

Scan code => Code of key pressed on
key board

* Scan code comes 2 time
1 for press, 1 for release

* Lower 7 bit contain key no.

CLI => Clear interrupt flag

Interrupt hooking => placing segment
and offset

Port 37A => Control port

Int 80 => Multitasking kernal interrupt

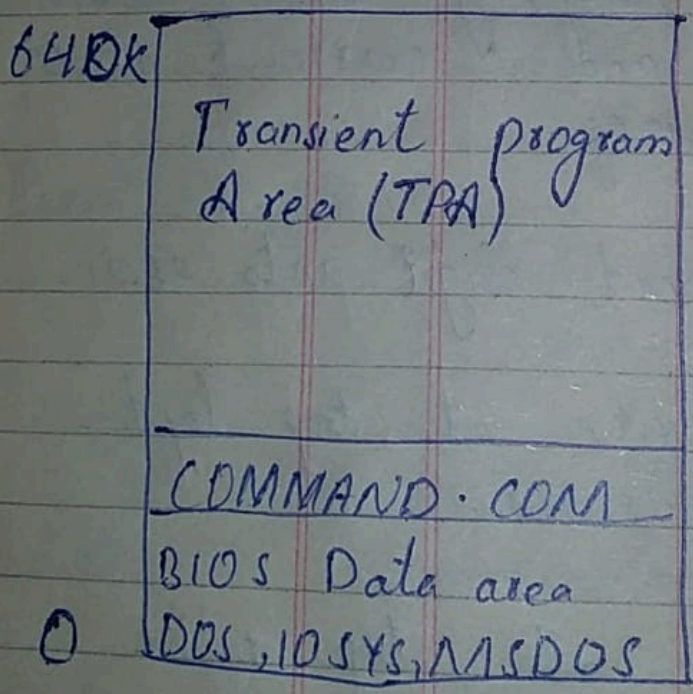
Date	PARTICULARS	Folio	Receipts	Issued	Balance	Rem

Service 2a => left shift press

" aa => left shift release

" 36 => right shift press

" b6 => right shift release



TSR => Terminate and stay resident.

Minimum

QUANTITY

Date | PARTICULARS | Folio | Receipts | Issued | Balance | Rem

Maxim
Minim
Date

DOS service 4C ⇒ allocating and deallocating memory in TPA.

Service 31 ⇒ terminate a program

Paragraph ⇒ 16 bytes (reserve in DX)

mem/c ⇒ ~~load~~ display all loaded drivers and current state of memory

service 30 ⇒ convert digit into ascii

(es:0) ⇒ write at top left of screen

[es:158] ⇒ write at top right of screen.

[es:8*4+2] ⇒ hook timer interrupt

LPT 1 => 378 to 37A ports

LPT 2 => 278 - 27A ports

Port 37A + 4 bit => enable IRQ7 + Ackp.

* AC current is 50 Hz sine wave.

flags not used for mathematical operations } Direction flag
Interrupt flag
trap flag.

Debugger used => Int 0, int 3
↓
Single byte opcode

INTO is used in multitasking

PCB => process control block
* Here all registers of task are stores.

Int 8 => called scheduler
* It saves and restores registers

Context switch => Whole event of multitasking

Minimum

Date	PARTICULARS	Folio	Receipts	Issued	Balance	Remarks

PIT
⇒ Programmable interval time

* Chiped name 8254, frequency 1.19318 MHz

* time stick ⇒ 18.2 times / sec

55 ms interval b/w two time ticks
and 8 is general after every interval

Parallel port
⇒ Two views
↳ Connector (interface)
↳ parallel port controller
(Process communicated with device)

⇒ Connector has 25 pin called DB-25

Data pin ⇒ pin 2 to pin 9

Pin 10/ACK ⇒ acknowledgement of data from printer

~~Pin~~ Ground pin ⇒ Pin 18-25

Date	PARTICULARS	Folio	QUANTITY Receipts	Issued	Balance	Remarks

Each sector on floppy disk holds 512 bytes

Standard floppy disk contain

⇒ 80 tracks

⇒ 18 sectors / track total 2880 sectors

⇒ 1440 KB data can store

⇒ Two heads to read data

Main units of DOS are in boot sector
head 0, track 0, sector 1

First FAT ⇒ head 0, track 0, sector 2

2nd FAT ⇒ head 0, track 0, sector 11

Root directory ⇒ head 1, track 0, sector 0

For storing data { head 0, track 1, sector 16
head 1, track 79, sector 18

Date	PARTICULARS	Folio	Receipts	Issued	Balance	Remarks

Int 80 => multi tasking kernal interrupt

Vedio services
- Graphic mode (320×200) mode D
- Text mode
  - 40×25 mode 0
  - 80×25 mode 2
  - 80×50 mode 3

* 8×16 font is stored in 16 bytes in ROM

0C07 => put pixel in white color

DOS video services
- More cooked
- higher level
- less control
- Easy routine task

first 446 bytes in hard disk contain MBR (

their 446 by partition enteries

2 byte signature

Directly ⇒ specially organized file

Int 21 services

⇓

* AH = 3Ch (create file)
* AH = 3Dh (open file)
* AH = 3Eh (close file)
* AH = 3Fh Read from file
* AH = 40h write to file
* AH = 41h Delete file
* AH = 42h Set file position
* AH = 4300h Get file attributes
* AH = 4301h set file ;

Minimum — QUANTITY

Date	PARTICULARS	Folio	Receipts	Issued	Balance	Remarks

AH = 48 h allocate memory

AH = 49 h free memory

AH = 4A h Resize memory block

A = 4B h Load AND/OR execute Program

~~Derive~~

Device Drivers

⇒ Extention of OS

* starts with header

15 bit ⇒ most important byte for checking char / block device.

Commands for drivers

- 0 to initialize
- 2 to build BIOS parameter
- 4 to read
- 8 to write

First 13 bytes are same for every command of request header

Maximum
Minimum

Date | PARTICULARS | Folio | QUANTITY: Receipts | Issued | Balance | Remarks

Serial / Parallel Port

=> Way of communication b/w two devices

Serial port
↳ bits are sent one by one
DB-9

Parallel port
↳ whole bits are sent at once

Serial port has DB-9 connection

1 Carrier detect CD
2 Received Data RD
3 Transmitted data TD
4 Data terminal ready DTR
5 Signal ground SG
6 Data set Ready DSR
7 Request to send RTS

Date PARTICULARS Folio Receipts

8 Clear to send CTS

9 Ring Indicator RI

Three wire connection for full duplex serial communication.

Standard formate RS232

Components of Serial communication

1 bit ⇒ start bit

5-8 ⇒ data bit

~~1~~ one to two zero bits ⇒ Stop bits

extra parity bit

brud rate ⇒ Duration of bit transmitting

Port parameter in AL consists

→ baud rate

→ Parity scheme

→ no of stop and data bit

Date PARTICULARS Folio Receipt

COM 1 ⇒ accessible at 3F8 - 3FF port

COM 2 ⇒ " " 2F8 - 2FF port

3FB is line control register

3FD is line status register

3FC is modem control register

8088 ⇒ 16 bit processor

Intel ⇒ 32 bit processor

32 bit address register can access 4GB memory.

FS and GS has no specific purpose

Segment descriptor describer

* base * limits * attributes

It describe actual base address

Remarks | Date | PARTICULARS | Folio | Receipts | Issued | Balance | Remarks

GDT (Global descriptor table)

LDT (Local " ")

RPL Requested privilege level
DPL Discriptor " "

13 bit index select descriptor in GDT.
it contain 8192 descriptors.

Each descriptor is of 8 byte

GDTR (48 bit register)
Here base and limit of GDT
is stored

VESA VBE 2.0 standard resolution mode.

VESA => Vedio electronics standards association

VBE => Video BIOS extention

Vedio services int 10

AX = 4F00h get super VGA Info

AX = 4F01h get super VGA mode info

AX = 4F02h set vesa video mode

VESA defined mode

4117 / 1024 × 768 mode

⇒ 16 bit color

⇒ linear buffer frame

⇒ 16 bits organized for each pixel as

5 : 6 : 5

↓ blue (32 shade)

↘ Red 32 shades.

↓ green (64 shades)

* 64 K total possible colors

Calling Convention (Behaviour of calling function.

- C calling
- Pascal calling

HLL ⇒ higher level languages

Date	PARTICULARS	Folio	Receipts	Issued	Balance	Rem

Naming Convention

- C
  - underscore to function
- Pascal
  - name in uppercase :

Pushing / Passing arguments

In C
  Right most pused first

In pascal
  left most :- "

Removing Parameter

In C
  caller removes

In pascal
  Calee removes .

Remarks | Date | PARTICULARS | Folio

Sun SPARC belongs to RISC

RISC => Reduced instruction set computer

CISC => Complex " "

SPARC => Scalable processor architecture

Data type

↳ byte, halfword, word, double word and quad word.

SPARC contain 32 bit instruction size.

Registers in SPARC

* 8 Global registers (r0 - r7)
* 8 In registers (r8 - r15)
* 8 local registers (r16 - r23)
* 8 out registers (r24 - r31)

In spark => Register window (24 registers)

~~CAT~~

Popular Processors

* Motorolla 68k are popular in high performance embedded application.

* Sun, sparc popular in high end enterprise services.

Motorolla 68k has 1623 bit genral purpose register.

named from A0-A7 and D0-D7

A0-A7 hold address in indirect memory.

Addressing is by inidicated by parenthesis.

68k has paged memory model.

In 68k is formal of instruction

operation source deslination

68k data types

byte, word, long word, quad word

Maximum 88 ARTICLE Rate

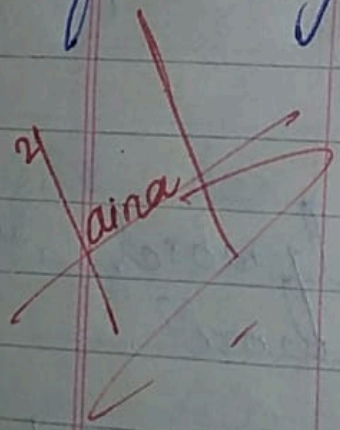
Minimum

ate	PARTICULARS	Folio	QUANTITY Receipts	Issued	Balance	Rema

CWP (current window Pointer)
It is a register

Spare contan 2 source
1 destination

allow more flexibily in writing
Program.



Date PARTICULARS Folio Receipts

Some kinds of interrupts

Int 10 ⇒ Vedio services

Int 16 ⇒ keyboard services

Int 17 ⇒ parallel port services

Int 21 ⇒ Single entry point in DOS

Int 10 service 13 ⇒ print a string

Int 16 service 1 ⇒ blocking service

Int 71 ⇒ blue on white blinking

int F4 ⇒ red on white blinking

int pin ⇒ generate intrrupts

Int 60 ⇒ Read a char

Service 0 ⇒ key stroke

Int 21 service 31 ⇒ Termination of program.

Int 20 ⇒ send EOI to PIC.